

Vinsea A V Singh

PhD Candidate · Cognitive and Computational Neuroscience

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Education

National Brain Research Centre (NBRC) Manesar

PhD Neuroscience (*Submitted*)

- Thesis: The Brain Before Perception: Trial-Wise Prestimulus Neural Dynamics Shape Multisensory Perception

Haryana, India
August 2020 - Present

National Brain Research Centre (NBRC) Manesar

MSc Neuroscience

- Dissertation: Characterizing variability of audiovisual speech perception based on prestimulus oscillatory features of electrophysiological brain signals [Link]

Haryana, India
August 2018 - July 2020

St. Xavier's College, Mumbai

BSc Zoology and Biochemistry

Mumbai, India
June 2014 - March 2017

Publications

Published

Singh, V. A., Roy, D., and Banerjee, A. (2026). Prestimulus EEG Studies in Multisensory Perception. In *Electroencephalogram Signals and Cognition* (pp. 49-73), edited by Hema A. Murthy, Shrikanth Narayanan, and Mriganka Sur. Sriranga Digital Software Technologies Pvt. Ltd. [Link]

Singh, V. A., Kumar, V. G., Banerjee, A., and Roy, D. (2025). Prestimulus Periodic and Aperiodic Neural Activity Shapes McGurk Perception. *eNeuro*, 12(10). <https://doi.org/10.1523/ENEURO.0431-24.2025>

Singh, V. A., Kumar, V. G., Banerjee, A., and Roy, D. (2022). Characterizing Variability of Audiovisual Speech Perception Based on Periodic and Aperiodic Features of Prestimulus Brain Activity. *bioRxiv*. 10.1101/2022.01.20.477172v1

In Prep

**contributing equally*

Singh, V. A., Kumar, V. G., Banerjee, A., and Roy, D. (*in prep*). Directional Connectivity in Prestimulus Large-Scale Functional Networks Underpins Multisensory Perception.

Bhunia, D.*, **Singh, V. A.***, and Roy, D. (*in prep*). A Thalamocortical Neural Field Model Better Elucidate Dynamics of Multisensory Perception. (*Co-leading analysis and drafting*)

Deb, S., Bhunia, D., **Singh, V. A.**, Muralidharan, V., and Roy, D. (*in prep*). Modulation of the Speed of Intrinsic Alpha Frequency (IAF) with tACS Modulates Variability of Illusory Speech Perception. (*Analysis and drafting*)

Singh, V. A.*, P.V*, Jain P*, Singhal T*, Singh P*, *et al.*, Banerjee A., (*in prep*) Multimodal MRI Assessment of Accelerated Cognitive Decline in Older Rural Indian Population. (*Co-leading analysis and drafting*)

Research Experience

“Advancing Approaches for Dementia Associated Research (AADAR)”

Dr. Arpan Banerjee (Supervisor) & INCLEN Trust (Collaborator)

NBRC
April 2025 - Present

- Co-leading SIEMENS PRISMA 3T-MRI acquisition, quality control (MRIQC), data curation (BIDS), and MRI/fMRI data pre-processing and SC-FC analysis for a demographically underrepresented older-age cohort in Northern India (To date: N = 359, 40 dementia, 123 MCI, and 196 normal; ages: 55-101), including T1-MPRAGE, T2-SPACE, T2-GRE, FLAIR, DTI, rs-fMRI, and PASL.
- Co-authored a continuation grant proposal (under review) for the AADAR dementia project in collaboration with my PhD supervisor. Drafted the core methodology and scientific rationale (see *Grants section*).

“Modulation of the Speed of Intrinsic Alpha Frequency (IAF) with tACS Modulates Variability of Illusory Speech Perception”

IIT Jodhpur & NBRC
2024-Present

Advisors: Dr. Dipanjan Roy, Dr. Vignesh Muralidharan

- Contributed to McGurk perception experiment design: 64-Channel EEG acquisition planning, and tACS stimulation protocol design; Co-led behavioral data analysis (rm-ANOVA, GLM, Bayesian) and IAF estimation (EEG pre-processing and power spectrum analysis); Result interpretation in collaboration with co-first authors.

“A Thalamocortical Neural Field Model Better Elucidates the Dynamics of Multisensory Perception”		IIT Jodhpur & NBRC
Advisor: Dr. Dipanjan Roy		2024-Present
Contributed to empirical data analysis of 3 datasets (McGurk-EEG, SIFI-EEG, & McGurk-iEEG) and design of the thalamocortical neural-field model, result interpretation, and paper writing in collaboration with first author (Co-author role).		
“Comparitive mapping of common mental disorders (CMD) over lifespan”		NBRC
Dr. Arpan Banerjee (Supervisor)		2022-2024
Standardized simultaneous EEG-fMRI protocol and preprocessing pipeline for passive moving watching task on SIEMENS 3T PRISMA MRI machine with 64-Channel passive electrode EEG system.		
“The Brain Before Perception: Trial-Wise Prestimulus Neural Dynamics Shape Multisensory Perception”		NBRC
Advisors: Dr. Arpan Banerjee (Supervisor), Dr. Dipanjan Roy (Co-Supervisor)		2020-Present
My PhD thesis work wherein I have identified EEG spectral, cross-spectral, and functional connectivity patterns at sensor and source space that predict response to multisensory stimulus using mixed-effect regression models, Granger-Geweke Causality, and Bayesian inference.		
“Characterizing variability of audiovisual speech perception based on prestimulus oscillatory features of electrophysiological brain signals”		NBRC
Advisor: Dr. Dipanjan Roy (Supervisor)		2019-2020

Grants & Funding		
ICMR Intermediate Grant	Co-Contributor, Characterizing early diagnostic markers for progression into dementia from mild cognitive impairments (MCI) and healthy aging in rural monolingual populations: A multi-site study (PI: Dr. Arpan Banerjee)	Under Review (2026)- Passed 2nd Round

Research Skills & Technical Expertise	
Experiment designing	Presentation software and Psychopy.
MRI / fMRI 3T acquisition	T1-MPRAGE, T2-SPACE, T2-GRE, FLAIR, PASL, DTI, rs-fMRI, BIDS curation
EEG acquisition & analysis	Active 64-channel EEG acquisition, preprocessing, artifact rejection, time–frequency analysis, power-spectrum, aperiodic fitting, coherence, PLV, Cross-frequency coupling, source estimation & time-series reconstruction, & Atlas-based parcellations
Simultaneous EEG–fMRI	Acquisition and analysis: synchronization and hardware integration, MRI-related EEG artifact rejection (gradient and pulse), multimodal data integration.
Statistical analysis	Mixed-effects regression models (R, python), rmANOVA (MATLAB, R), Bayesian statistics (R), correlation (R), LASSO regression (R, python), Granger causality (MATLAB), permutation testing (MATLAB, R, python), Drift-diffusion model (python).
Machine learning	GLMs (linear, logistic), Classification (SVM, k-NN), RSA, Dimensionality reduction (PCA, t-SNE), Graph-based Network, cross-validation; Familiarity with: predictive neural decoding and encoding model
Programming	MATLAB, Python, and R.
Toolboxes	BrainVision RecView, EEGLAB, ERPLAB, GEDAI, FieldTrip, MNE, Chronux, Specparam (FOOOF), NBS, fMRIB-EEGLAB, fMRIPrep, MRIQC, Nilearn.

Awards, Achievements & Fellowships		
2025	NBRC Travel Grant to attend OHBM 2025, Brisbane, National Brain Research Centre (NBRC), Manesar	Rs.75,000

2024	EMBO Travel Grant , Barcelona Summer School for Advanced Modeling of Behavior (BAMBI)	1,000 euros
	Journal of Cognitive Neuroscience (JoCN) Travel Award , Cognitive Neuroscience Society (CNS), Toronto, Canada	3,000 USD
2023	OHBM 2023 Travel Award , Organization for Human Brain Mapping (OHBM)	~4,000 USD
2020-Present	NBRC PhD in Neuroscience Fellowship , National Brain Research Centre, Manesar	
2018-2020	NBRC Masters in Neuroscience Fellowship , National Brain Research Centre, Manesar	
2017	Certificate of Merit for pursuing The Honors Degree in Zoology , St. Xavier's College, Mumbai	
2016	Summer Research Fellow (SRF) , Indian Academy of Sciences (IAS)	

Presentations

Invited Talks

- Singh, V. A.**, Kumar, V. G., Banerjee, A., and Roy, D. 2026. Prestimulus large-scale directed connectivity patterns predispose multisensory perception. SfN 2026 Nanosymposium, session: *"Human Audiovisual Integration"*, DC, USA (*Upcoming*)
- March 2024. *Neuroscience: the past, present and the future*. Invited talk: National Science Day, Birla Balika Vidyapeeth (my alma mater), Pilani, Rajasthan, India.
- February 2022. *Role of prestimulus activity in characterizing variability in multisensory perception*. Invited talk: CogSci Seminar Series, hosted by Dr. Vishnu Sreekumar, International Institute of Information Technology (IIIT) Hyderabad, India.
- Singh, V. A.**, Kumar, V. G., Banerjee, A., and Roy, D. 2021. Characterizing Variability of Audiovisual Speech Perception Based on Prestimulus Oscillatory Features of Electrophysiological Brain Signals. Oral presentation: 7th Annual Conference of Cognitive Sciences (ACCS7) 2021, Online

Poster Presentations

* presenting author; * mentored postgraduate

- Singh, V. A.**, Kumar, V. G., Banerjee, A., and Roy, D. 2025. Prestimulus Whole-Brain Functional Connectivity Underpins McGurk Perception. Organization for Human Brain Mapping (OHBM) 2025, Brisbane, Australia. *Won a Travel Award*.
- Singh, V. A.**, Kumar, V. G., Banerjee, A., and Roy, D. 2024. Predicting Response to McGurk Effect Based on Periodic and Aperiodic Prestimulus EEG Activity. Cognitive Science Society (CogSci) 2024, Rotterdam, Netherlands.
- Singh, V. A.**, Kumar, V. G., Banerjee, A., and Roy, D. 2024. Predicting Response to McGurk Effect Based on Periodic and Aperiodic Prestimulus EEG Activity. Cognitive Neuroscience Society (CNS) 2024, Toronto, Canada. *Won a Travel Award*.
- Singh, V. A.**, Kumar, V. G., Banerjee, A., and Roy, D. 2023. Predicting Response to McGurk Effect Based on Periodic and Aperiodic Prestimulus EEG Activity. Organization for Human Brain Mapping (OHBM 2023), Montreal, Canada. *Won a Travel Award*.
- Singh, V. A.**, Kumar, V. G., Banerjee, A., and Roy, D. 2022. Characterizing Variability of Audiovisual Speech Perception Based on Prestimulus Oscillatory Features of Electrophysiological Brain Signals. 9th Annual Conference of Cognitive Sciences (ACCS9) 2021, New Delhi, India
- Singh, V. A.**, Kumar, V. G., Banerjee, A., and Roy, D. 2020. Characterizing Variability of Audiovisual Speech Perception Based on Prestimulus Oscillatory Features of Electrophysiological Brain Signals. Bernstein Conference 2020, Online.
- Deb, S.*+, **Singh, V. A.**, Muralidharan, V., and Roy, D. 2025. Modulation of the Speed of Intrinsic Alpha Frequency (IAF) with tACS Modulates Variability of Illusory Speech Perception. 12th Annual Conference of Cognitive Sciences (ACCS12) 2025, Allahabad, Uttar Pradesh, India.
- Bhunia, D.*+, **Singh, V. A.**, and Roy, D. 2026. Thalamocortical Neural Field Model for Inter-individual and Inter-Trial Variability in Multisensory Speech Perception. BrainDy'26, Udaipur, Rajasthan, India.

Teaching Experience

July 2026	Computational Neuroscience Course , TA, Neuromatch Academy (NMA) Summer School	Online
Fall 2023	Biostatistics Course , Graduate TA	NBRC, India
Spring 2023	Cognitive Neuroscience Course , Graduate TA	NBRC, India
July 2022	Computational Neuroscience Course , Project TA, NMA Summer School	Online

Mentoring

* mentored (under Co-PI supervision); * Lead mentor

- 2024-Present **Shreya Deb***, MS by Research, Indian Institute of Technology (IIT) Jodhpur
- August-September 2023 **Bharti Upparu***, Mentee, WISR (Women in STEM Research) mentorship program, *Guided on building a research career in Cognitive Psychology*
- Summer 2023 **Antarjot Kaur***, Summer Intern, Indian Institute of Technology (IIT) Jodhpur, *Won best internship award*
- Summer 2023 **Devanshi Singh***, Summer Intern, Indian Institute of Technology (IIT) Jodhpur
- Summer 2022 **Palak Sharma***, Summer Intern, Indian Institute of Technology (IIT) Jodhpur, *Won best internship award*

Other Research Experience

“Analysis of the Sex-Ratio and Body Length of *Carcharhinus limbatus* (Blacktip Shark) Over Varied Seasons”

Mumbai, India

Research Volunteer, Mumbai Shark Catches Monitoring Programme, Sassoon Dock

2015-2017

- Nature Conservation Foundation
- Assisted in data collection, compilation, and analysis of the blacktip shark catches that arrived at the Sassoon dock, Mumbai

“Female Mate Choice Based on Male Age: Are older or younger males better mates in the primitively eusocial wasp *Ropalidia marginata* (paper wasps)?”

Bangalore, India

Supervisor: Professor Raghavendra Gadagkar

May-June 2016

- Centre for Ecological Sciences (CES), Indian Institute of Science Bangalore (IISc)
- Studied the mate selection behavior in female paper wasps

Outreach & Professional Development

Service & Outreach

- January 2026 **Scientific Committee Member**, BraiNDy 2026, Udaipur
- December 2025 **Instructor for MRI and EEG hands-on**, Neuroimaging Conference - Clinical and Preclinical Perspectives 2025, National Brain Research Centre (NBRC)
- 2023-2024 **Fellow**, OHBM Student PostDoc Special Interest Group (OHBM SP-SIG) Fellowship Programme
- December 2023 **Volunteer**, Symposium on Brain, Behavior, & Society, National Brain Research Centre (NBRC)
- August 2023 **Mentor**, WISR - Women in STEM Research Mentorship Program
- July 2023 **TA**, “The Brain Workshop” organized by the Artificial Intelligence and Data Science (AIDE) School of Indian Institute of Technology (IIT), Jodhpur
- July 2022 **Project TA**, Computational Neuroscience, Neuromatch Academy (NMA) Online Summer School
- 2022-2023 **Committee Member**, NBRC Journal Club
- 2022, 2023 **Scientific Representative of NBRC at MegaScience Expo**, Indian International Science Fest (IISF) 2022 and IISF 2023

Summer Schools

Barcelona Summer School for Advanced Modeling of Behavior (BAMB!), July 2024, Barcelona, Spain

Neuromatch Academy (NMA), Computational Neuroscience, July 2020, Online.

Conference Reviewer

NeuroAI 2026, OpenReview.net

Work Experience

Research Consultant at Moment (now called The Biohackers Co.), April 2022 - December 2022

References

Dr. Arpan Banerjee, Scientist VI & Professor, NBRC, arpan@nbrc.ac.in, +91-124-2845410

Dr. Dipanjan Roy, Professor, IIT Jodhpur, droy@iitj.ac.in, +91-291-2801757